

Now the value of `x` will start at 1 but never change. As a result, the conditional test `x <= 5` will always evaluate to `True` and the `while` loop will run forever, printing a series of 1s, like this:

```
1
1
1
1
--snip--
```

Every programmer accidentally writes an infinite `while` loop from time to time, especially when a program's loops have subtle exit conditions. If your program gets stuck in an infinite loop, press `CTRL-C` or just close the terminal window displaying your program's output.

To avoid writing infinite loops, test every `while` loop and make sure the loop stops when you expect it to. If you want your program to end when the user enters a certain input value, run the program and enter that value. If the program doesn't end, scrutinize the way your program handles the value that should cause the loop to exit. Make sure at least one part of the program can make the loop's condition `False` or cause it to reach a `break` statement.

NOTE

VS Code, like many editors, displays output in an embedded terminal window. To cancel an infinite loop, make sure you click in the output area of the editor before pressing `CTRL-C`.

TRY IT YOURSELF

7-4. Pizza Toppings: Write a loop that prompts the user to enter a series of pizza toppings until they enter a 'quit' value. As they enter each topping, print a message saying you'll add that topping to their pizza.

7-5. Movie Tickets: A movie theater charges different ticket prices depending on a person's age. If a person is under the age of 3, the ticket is free; if they are between 3 and 12, the ticket is \$10; and if they are over age 12, the ticket is \$15. Write a loop in which you ask users their age, and then tell them the cost of their movie ticket.

7-6. Three Exits: Write different versions of either Exercise 7-4 or 7-5 that do each of the following at least once:

- Use a conditional test in the `while` statement to stop the loop.
- Use an active variable to control how long the loop runs.
- Use a `break` statement to exit the loop when the user enters a 'quit' value.

7-7. Infinity: Write a loop that never ends, and run it. (To end the loop, press `CTRL-C` or close the window displaying the output.)